Medical Imaging Dataset Exploration

Dataset 1: NIH Chest X-ray14

https://www.kaggle.com/datasets/nih-chest-xrays/data?utm_source=

Type of Imaging Data: Chest X ray

Number of Images: 112,120

Available Classes:
Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Edema, Fibrosis, Emphysema, Pleural Thickening, Hernia, No Finding.

Dataset Imbalance:
The dataset is imbalanced because normal images and some disease categories have significantly more samples than rare diseases such as Hiatal Hernia.

Challenges:
• Label noise due to automatic annotation methods.
• Multiple diseases can occur in the same image.
• Limited localization information.

Brief Summary:
The NIH Chest X ray14 dataset is a large scale chest radiography dataset used for automated disease detection. It is widely employed in medical imaging research and deep learning applications.

Dataset 2: HAM10000 Skin Lesion Dataset

https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000?utm_source=

Type of Imaging Data: Skin lesion images.

Number of Images: 10,015

Available Classes:
Melanoma, Melanocytic Nevi, Basal Cell Carcinoma, Benign Keratosis, Dermatofibroma, Vascular Lesions, and Actinic Keratoses.

Dataset Imbalance:
The Melanocytic Nevi class contains the majority of images, while several disease classes have relatively few samples.

Challenges:
• Class imbalance.
• Hair and skin artifacts.
• Variations in image quality and illumination.

Brief Summary:
HAM10000 is a popular skin lesion dataset used for skin cancer classification research. It supports the development of AI models that assist in early detection of melanoma and other dermatological conditions.